

5 Work, energy and power

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Work is energy transfer by a force through displacement in its direction: $W = Fs \cos \theta$. Kinetic energy is $\frac{1}{2}mv^2$; gravitational energy change near Earth is $mg\Delta h$. Power measures energy transferred per unit time. For a real process, conservation applies to all stores even when useful efficiency is below 100%.

Worked example: follow the reasoning

A 2 kg mass rises 5 m: $\Delta E = 2 \times 9.81 \times 5 = 98.1$ J. In 4 s the minimum input power is 24.5 W if losses are negligible.

Quick check · multiple choice

A Which expression gives kinetic energy?

[1]

A $\frac{1}{2}mv^2$

B mv

C mgh

D F/t

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A motor lifts a 30 kg mass through 2.0 m in 5.0 s while drawing 150 W. Calculate efficiency.

[3]

Final answer: _____

Answers and explanation

Quick check: A. $\frac{1}{2}mv^2$

Question 1 [3 marks]

Useful power = $mgh/t = 30 \times 9.81 \times 2/5 = 117.7 \text{ W}$ [2]. Efficiency = $117.7/150 = 0.785$, or 78.5% [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.